

P4 v2: a global lower bound on $n(r, U)$ from summing `thm:arc`'s local pole rate

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Labels: PROVED means a complete argument, given the inputs it cites (here, given `thm:arc/cor:arcUV` of paper2a, which are themselves conditional on Theorems `thm:model` and `thm:RJ`). VERIFIED means high-precision floating point checked against paper2a's own reported numbers, not a fresh interval certificate. HEURISTIC means an argument without full error control. This note continues `P4/P4_note.tex` ("P4v1"), which identified two missing ingredients for a Nevanlinna-theoretic resolution of P4 (is U D-algebraic): (a) a quantitative global pole-counting rate $n(r, U)$ as $r \rightarrow 1^-$, summed from the local rate `thm:arc` gives near each fixed root of unity; (b) a disc-analogue of the Bank–Kaufman/Steinmetz counting theorem. P4v1 explicitly did not carry out the summation in (a). This note carries it out.

Correction (post-review, Reviewer AN): the headline claim is NOT established

The local-rate formula (1) below, and everything built on it (§2–3, Proposition 2), is wrong as a claim about general ζ . The formula $t_j^* = (L + i\pi)/(2j + 3/2) + O(t)$ was copied from `prop:minusone` (paper2a, the *special case* $\zeta = -1$), not from the general theorem that actually backs `thm:arc` (`P1f_lemma.tex`, the general saddle/Rouché theorem with $\Delta V = (\pi i s/N)(L_n + i\pi s/N)$, $s \in \{1, 2\}$). The general ΔV carries an extra factor of order $1/N$ that the $\zeta = -1$ formula (a fixed small case, $N = 2$) hides, so the correct leading behavior is $1 - |q_j^*(\zeta)| \sim C(\zeta, N)/j$ with $C(\zeta, N) = O(1/N)$, **not** $\ell(\zeta)/(2j)$. Checking $\zeta = i$ (`prop:qi`) independently gives yet a third, different offset ($2j + 1$, from $m = -j - 1$), confirming the coefficient is genuinely case-specific and not the single formula claimed below. The numerical "corroboration" at $\zeta = i$ is correspondingly not meaningful: at $j = 1, 2$ the wrong-for-this-case formula (with offset $3/2$) happens to fit slightly *better* than $\zeta = i$'s own correct offset ($2j + 1$, predicting ratio $5/3 = 1.667$, a worse 3.8% match) – this is $O(1/j)$ -correction noise, not evidence for the general shape.

What survives, honestly: the *qualitative* diagnosis that P4v1's back-of-envelope $\log(1/(1 - r))$ -per- ζ guess conflates a harmonic-sum question with a threshold-count question, and that the correct per- ζ count is linear (not logarithmic) in $1/(1 - r)$ whatever its exact coefficient turns out to be – and the observation that a diagonal argument over finitely many fixed ζ at a time needs no uniform-in- ζ control, only individually-finite thresholds. Both are modest, correct points. The specific formula (1), Lemma 1's use of it, and Proposition 2's conclusion are **not established** by this note as written; a correct version would need the general ΔV -based rate re-derived from `P1f_lemma.tex` directly, plus (per Reviewer AN) an honest count of how many admissible (a, N) contribute a non-negligible coefficient once the true $O(1/N)$ dependence is accounted for – not

attempted here. **P4 remains exactly as open as P4v1 left it; ingredient (a) is still not carried out.**

Summary of outcome (as originally written; superseded by the correction above)

Claimed result, NOT established (see correction above): for every $A > 0$,

$$n(r, U) \geq \frac{A}{1-r} \quad \text{for all } r \text{ sufficiently close to } 1,$$

i.e. $\liminf_{r \rightarrow 1^-} (1-r) n(r, U) = +\infty$. The argument below is retained for the record and because its qualitative log-vs-linear point and its diagonal-argument shape are genuine, but its quantitative content rests on the incorrect formula (1) and must not be cited as a proved strengthening of P4v1.

1 The local rate, extracted precisely

From `paper2a.tex`, `thm:arc` (§?? there) and its corollary `cor:arcUV`: fix $\zeta = e^{2\pi ia/N}$, $\gcd(a, N) = 1$, admissible in the sense of `cor:arcUV` ($N \geq 16$, $\frac{a}{N} \in [\frac{1}{6}, \frac{5}{6}]$; wider ranges are available via P1h/P1i but are not needed below). Write

$$\ell = \ell(\zeta) := \log|2(1-\zeta)|, \quad L = L(\zeta) := \text{Log}(-2(1-\zeta)),$$

so $\text{Re } L = \ell$. The proof of `thm:arc` produces, for each such ζ , an explicit sequence of poles $q_j^*(\zeta) = \zeta e^{-t_j^*}$ of G_0^T (hence, by `cor:arcUV`, of U and V after transfer through $x = \pm \sqrt{q_j^*}$), valid for all j past some finite, ζ -dependent threshold $j_0(\zeta)$ (“the index beyond which q_j^* is a pole ... is not made explicit”, `rem:arcUV`), with

$$t_j^* = \frac{L + i\pi}{2j + 3/2} + O(t) \quad (j \rightarrow \infty), \quad \text{so} \quad \text{Re}(t_j^*) = \frac{\ell}{2j + 3/2} (1 + o(1)).$$

Since $|q_j^*| = e^{-\text{Re}(t_j^*)}$, and $\ell > 0$ on the arc (required for the resummation to apply at all, `thm:arc`’s statement), $\text{Re}(t_j^*) \downarrow 0$ monotonically for large j , i.e. $|q_j^*(\zeta)|$ *increases to 1 as $j \rightarrow \infty$* , and

$$1 - |q_j^*(\zeta)| = \text{Re}(t_j^*) (1 + o(1)) = \frac{\ell(\zeta)}{2j} (1 + o(1)) \quad (j \rightarrow \infty). \quad (1)$$

Numerical corroboration (VERIFIED against paper2a’s own reported values, no new computation). At $\zeta = i$ ($N = 4$, outside the $N \geq 16$ arc but governed by the same mechanism, Corollary `cor:Ui`), paper2a reports $q_1^* = 0.1404 + 0.8293i$, $q_2^* = 0.0802 + 0.8974i$, so $1 - |q_1^*| = 1 - 0.8411 = 0.1589$ and $1 - |q_2^*| = 1 - 0.9010 = 0.0990$. Equation (1) predicts the ratio $(1 - |q_1^*|)/(1 - |q_2^*|) \approx (2 \cdot 2 + 3/2)/(2 \cdot 1 + 3/2) = 5.5/3.5 = 1.571$; the reported values give $0.1589/0.0990 = 1.605$, a 2% match consistent with the stated $O(t)$ correction at these small j . This is a sanity check on (1)’s shape, not a new certificate.

2 Inverting the local rate: linear, not logarithmic, per- ζ count

Fix an admissible ζ and $r \in (0, 1)$. Using $1 - r \leq -\log r \leq (1 - r)/r$, the two notions of “close to the boundary” agree to leading order as $r \rightarrow 1^-$, so I write $-\log r = 1 - r + O((1 - r)^2)$ and suppress the difference below. By (1), $|q_j^*(\zeta)| \leq r$ iff $\operatorname{Re}(t_j^*) \geq -\log r$, i.e. (for j past the threshold where (1) is valid)

$$\frac{\ell(\zeta)}{2j + 3/2}(1 + o(1)) \geq 1 - r + O((1 - r)^2) \iff j \leq \frac{\ell(\zeta)}{2(1 - r)}(1 + o(1)) - \frac{3}{4}.$$

So the number of $j \geq j_0(\zeta)$ with $|q_j^*(\zeta)| \leq r$ is

$$n_\zeta(r) := \#\{j \geq j_0(\zeta) : |q_j^*(\zeta)| \leq r\} = \frac{\ell(\zeta)}{2} \cdot \frac{1}{1 - r} - j_0(\zeta) + o\left(\frac{1}{1 - r}\right) \quad (r \rightarrow 1^-). \quad (2)$$

This is **linear** in $1/(1 - r)$, with an explicit leading coefficient $\ell(\zeta)/2$, not the $\log(1/(1 - r))$ shape guessed in P4v1 §5(a). (The log intuition would be the right one if the question were “how large is $\sum_j [\text{something } 1/j]$ ” – a harmonic-sum question – but the question here is “how many terms of a sequence decaying like $1/j$ exceed a threshold $1 - r$ ”, which is answered by inverting $\ell/(2j) = 1 - r$ directly, giving a term count linear in $1/(1 - r)$, not a harmonic sum.) This is the concrete correction to P4v1’s back-of-envelope estimate that carrying out the summation actually produces.

3 Diagonal summation over finitely many roots of unity

Lemma 1. *There is a constant $\ell_{\min} > 0$ and infinitely many pairwise distinct admissible ζ (in the sense of §1) with $\ell(\zeta) \geq \ell_{\min}$.*

Proof. $\ell(\zeta) = \log |2(1 - \zeta)|$ is continuous and strictly positive on the closed arc $\frac{a}{N} \in [\frac{1}{6}, \frac{5}{6}]$ minus $\{a/N = \frac{1}{2}\}$... in fact it is strictly positive and continuous on all of $\arg \zeta \in [\frac{\pi}{3}, \frac{5\pi}{3}] \setminus \{0\}$, in particular on the compact sub-arc $\arg \zeta \in [\frac{\pi}{3} + \delta, \pi - \delta]$ for any small fixed $\delta > 0$, hence bounded below there by some $\ell_{\min}(\delta) > 0$. For any prime $N \geq 16$ the point $a = \lfloor N/2 \rfloor$ satisfies $\gcd(a, N) = 1$ automatically and $a/N \rightarrow \frac{1}{2}$ as $N \rightarrow \infty$, so for N large enough a/N lies in $[\frac{1}{6} + \delta', \frac{5}{6} - \delta']$ for any fixed $\delta' > 0$; there are infinitely many primes, giving infinitely many distinct admissible $\zeta_N = e^{2\pi i a/N}$, pairwise distinct since they have pairwise distinct arguments in $(0, 2\pi)$, with $\ell(\zeta_N)$ bounded below uniformly (by continuity, $\ell(\zeta_N) \rightarrow \ell(-1) = \log 4$ as $N \rightarrow \infty$ along this family, and is bounded below on the whole compact range visited). $\square$

Proposition 2 (main result of this note). *Assume `thm:model` and `thm:RJ` (so that `cor:arcUV` holds). Then $\liminf_{r \rightarrow 1^-} (1 - r) n(r, U) = +\infty$, where $n(r, U)$ counts poles of U in $|x| \leq r$ with multiplicity.*

Proof. Fix $A > 0$. By Lemma 1, choose finitely many distinct admissible $\zeta_1, \dots, \zeta_K$ (all with $N \geq 16$) with $\sum_{k=1}^K \ell(\zeta_k)/2 > A$; this is possible for K large enough since $\ell(\zeta_k) \geq \ell_{\min} > 0$ for infinitely many of them. Each ζ_k has a finite threshold $j_0(\zeta_k)$ (`thm:arc`’s proof; not numerically given, but finite). By (2), for each fixed k ,

$$n_{\zeta_k}(r) = \frac{\ell(\zeta_k)}{2} \cdot \frac{1}{1 - r} - j_0(\zeta_k) + o\left(\frac{1}{1 - r}\right) \quad (r \rightarrow 1^-).$$

Since K is fixed, summing these K (finitely many) asymptotics is legitimate term-by-term:

$$\sum_{k=1}^K n_{\zeta_k}(r) = \left(\sum_{k=1}^K \frac{\ell(\zeta_k)}{2} \right) \frac{1}{1-r} - \sum_{k=1}^K j_0(\zeta_k) + o\left(\frac{1}{1-r}\right) > \frac{A}{1-r} \quad \text{for } r \text{ close enough to 1 (depending on } K, \text{ hence on } A)$$

The poles counted at distinct ζ_k are, for r close enough to 1, genuinely distinct points of $\{|x| \leq r\}$: along the tie ray $\arg t = \theta^*(\zeta_k)$, $q_j^*(\zeta_k) \rightarrow \zeta_k$ in *both* modulus and argument as $j \rightarrow \infty$ (the perturbation from ζ_k has size $|t_j^*| = O(1/j)$ in every direction, not just radially), so for fixed distinct $\zeta_1, \dots, \zeta_K$ the corresponding pole clusters live in disjoint shrinking neighbourhoods of $\zeta_1, \dots, \zeta_K$ once j (equivalently r) is large enough; no double counting occurs. Transferring through $x = \pm \sqrt{q_j^*}$ and complex conjugation (`cor:arcUV`) multiplies every count by an absolute constant (≤ 4) and does not affect distinctness for the same reason (the four points $\pm \sqrt{q_j^*}, \pm \sqrt{\overline{q_j^*}}$ cluster near the four fixed points $\pm \sqrt{\zeta_k}, \pm \sqrt{\overline{\zeta_k}}$, distinct across distinct k once close enough). Hence $n(r, U) \geq \sum_{k=1}^K n_{\zeta_k}(r) > A/(1-r)$ for r close enough to 1. Since $A > 0$ was arbitrary, $\liminf_{r \rightarrow 1^-} (1-r)n(r, U) = +\infty$. $\square$

Remark 3 (why no uniform constants are needed here, and where they would be needed). The proof above needs, for each *fixed* K , only the K individual (finite, if unspecified) thresholds $j_0(\zeta_1), \dots, j_0(\zeta_K)$ – a quantifier order (“for all A there is $r_0(A)$ ”) that lets K , and hence the finite list of thresholds used, depend on A freely. This is why P4v1’s concern about uncontrolled implied constants does not block Proposition 2: no uniformity *across* ζ is used, only finitely-many, one-at-a-time applications of `thm:arc`. Uniformity across ζ would only be needed to let $K = K(r) \rightarrow \infty$ as $r \rightarrow 1$ (i.e. to use more and more roots of unity as r approaches 1, rather than a fixed finite list), which is exactly what would be needed to push the bound from “faster than every fixed multiple of $1/(1-r)$ ” to “faster than every fixed power of $1/(1-r)$ ” (superpolynomial) – see §5.

4 Comparison with what a disc Bank–Kaufman/Steinmetz bound would need

P4v1 §3 already flags that the plane-case Malmquist/Steinmetz/Bank–Kaufman theorems bound the Nevanlinna characteristic $T(r, y)$ (equivalently, via the first main theorem, $N(r, y) = \int_0^r n(t, y)/t dt$, hence $n(r, y)$ up to the usual logarithmic-derivative slack) of a transcendental meromorphic solution of a fixed algebraic ODE with polynomial/rational coefficients, in terms of the order k and degree d of the equation and the growth of its coefficients; for coefficients of bounded degree the resulting bound on $T(r, y)$ is of *finite order* in r (a fixed power of r) as $r \rightarrow \infty$ on $\mathbb{C}$. The natural disc analogue (Tsuji-type; Bank’s disc papers, per `rem:disc` of P4v1, cited from memory and not independently re-verified here either) should bound $T(r, y)$, as $r \rightarrow 1^-$, by a fixed power of $1/(1-r)$ – i.e. “finite order” becomes *polynomial growth in* $1/(1-r)$, for coefficients bounded by a fixed power of $1/(1-r)$ as well (rational functions of x alone have coefficients bounded near $|x| = 1$ unless they have poles exactly on the unit circle, which would need separate handling; assume for this comparison coefficients holomorphic on $\overline{D}$, the case relevant to ruling out a fixed ODE with, e.g., polynomial coefficients).

What Proposition 2 gives, and what it does not give, against this. Proposition 2 shows $n(r, U)$ is not $O(1/(1-r))$ for any constant – it beats every *linear-order-1* bound in $1/(1-r)$. If the disc Bank–Kaufman analogue for a fixed order- k , degree- d ODE gave a bound of the shape $n(r, y) = O((1-r)^{-\rho(k,d)})$ with $\rho(k, d) = 1$ for the simplest (e.g. Riccati, $k = 1, d = 2$) case,

Proposition 2 would already rule out *that specific* equation class. But it does *not* rule out equations with $\rho(k, d) > 1$ (which presumably exist for higher k, d , exactly mirroring how in the plane case higher-order algebraic ODEs admit solutions of higher finite order): to rule out *every* (k, d) simultaneously one needs $n(r, U)$ to beat $(1 - r)^{-\rho}$ for *every* fixed ρ , i.e. superpolynomial growth in $1/(1 - r)$, which is a strictly stronger statement than Proposition 2 and is **not established here**.

5 Diagnosis: what closing the superpolynomial gap needs

Pushing §3’s argument from “beats every constant multiple of $1/(1 - r)$ ” to “beats every power of $1/(1 - r)$ ” means letting the number of roots of unity used, K , grow as $r \rightarrow 1^-$ (e.g. $K(r) \sim (1 - r)^{-\rho}$ for growing ρ), and this needs, uniformly over the $K(r)$ many ζ ’s used simultaneously:

1. A bound on $j_0(\zeta)$ (the threshold past which (1) is valid) as a function of N (or of $\ell(\zeta)$), rather than the qualitative “eventually true” statement `thm:arc`’s proof gives. `rem:arcUV` states explicitly that this index “is not made explicit”.
2. A bound on how many admissible $\zeta = e(a/N)$ with $N \leq N_{\max}(r)$ exist, for $N_{\max}(r)$ tied to $1 - r$ – this part is elementary number theory (Euler-phi counting of reduced fractions in an interval, $\Theta(N_{\max}^2)$ many with $N \leq N_{\max}$) and is *not* an obstruction.
3. Crucially, the hazard-induction certificates behind Theorems M, M’, `main’` (`P1e_lemma.tex`, `P1g_lemma.tex`, `P1h_lemma.tex`) are stated as certified for N up to 10^5 by direct computation and by “a hand argument for $N > 10^5$ ” (`thm:arc`’s proof sketch, quoting `P1h_lemma.tex`); whether that hand argument gives an *explicit, uniform* lower bound on $|\omega|$ -margins / non-vanishing constants as $N \rightarrow \infty$ (as opposed to merely an existence statement for each large N) is not something this note verifies – it would need rereading `P1g_lemma.tex`/`P1h_lemma.tex` in full with this specific question (“is the $N \rightarrow \infty$ tail bound uniform and explicit?”) in mind, which was not done here.

If (1) and (3) can be answered with an explicit polynomial-in- N (or even just *any* explicit) bound on $j_0(\zeta)$, the diagonal argument of §3 upgrades mechanically: take $K(r) \rightarrow \infty$ at a controlled rate, and the leading coefficient $\sum \ell(\zeta_k)/2$ over $K(r) \sim (1 - r)^{-\rho}$ many terms (each $\ell(\zeta_k)$ bounded below, hence the sum growing at least linearly in $K(r)$) would give $n(r, U) \gtrsim (1 - r)^{-1-\rho} \cdot (1 - r)$ -type improvement – growing without bound in ρ , i.e. superpolynomial. This is plausible but is exactly the missing computation; it is not carried out here for lack of the explicit thresholds in (1)/(3), and is left as the next concrete step.

Weakest points

- **The whole note is conditional** on `thm:model`, `thm:RJ` (as `cor:arcUV` already is), and on the correctness of `thm:arc`’s proof sketch as quoted in paper2a (this note did not re-derive `thm:arc` from `P1f`/`P1h_lemma.tex` from scratch; it takes the stated asymptotic $t_j^* \sim (L + i\pi)/(2j + 3/2)$ at face value from paper2a’s text).
- §2’s inversion treats $1 - r \approx -\log r$ and drops the stated $O(t)/o(1)$ correction terms uniformly; this is standard but was not made into an explicit error bound (e.g. no claim of the form “for $r > 1 - \epsilon_0(\zeta)$ ”, only “eventually”). Making Proposition 2’s proof fully quantitative (an explicit $r_0(A)$) would require exactly the same threshold data flagged missing in §5.

- Proposition 2 is a genuinely new statement (not present in paper2a or in P4v1), but it is **weaker than what P4 needs**: superlinear-in-every-constant is not superpolynomial, and §4 shows superpolynomial is the actual threshold to beat every fixed-order, fixed-degree algebraic ODE via a Bank–Kaufman/Steinmetz-shaped bound (itself still uncited in citable disc form, ingredient (b), untouched here).
- Lemma 1’s use of primes N with $a = \lfloor N/2 \rfloor$ is one convenient witnessing family, chosen for a clean proof that $\gcd(a, N) = 1$; nothing hinges on primality beyond that convenience, and any infinite family of coprime pairs landing in a fixed compact sub-arc away from $\zeta = 1, -1$ would do equally well.
- The claim in §3’s proof that distinct- ζ_k pole clusters do not overlap for r close to 1 is argued qualitatively (tie-ray convergence in both modulus and argument); it was not turned into an explicit separation bound. For a fixed finite K this is not a real risk (compactness of the finite set $\{\zeta_1, \dots, \zeta_k\}$ suffices), but it is the same missing uniformity that would need to be resolved to let $K \rightarrow \infty$ with r , per §5.
- No computation (Rust, Arb, or otherwise) was run for this note beyond the arithmetic sanity check in §1, which reuses numbers already printed in paper2a; nothing here required new numerics, since the content is an asymptotic inversion and a diagonal summation argument, not a new numerical claim. If §5’s item (3) is pursued next, that would be the first place a fresh computation (checking whether `P1g_lemma.tex`’s $N \rightarrow \infty$ hand argument gives an explicit uniform bound) becomes relevant, and should be scoped as a targeted re-reading of `P1g_lemma.tex/P1h_lemma.tex`, not a new Rust enumeration.